

MASTERCARD

VISA

ENTRUST KEY MANAGER

EMV

nShield HSM

The Complete Guide to Integrating Mastercard & Visa into Entrust Infrastructure

A hands-on field guide for preparing and configuring the Entrust software stack for card issuance — token tables, issuer key pairs, CA requests, certificate imports, session keys, and EMV profiles.

prepared by MinThutaSawNaing · Card Personalization & Key Management Series

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ENTRUST PORTALS

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INTEGRATION STEPS

0

MASTERCARD FILES
(.CEF/.HEP/.SEP)

0

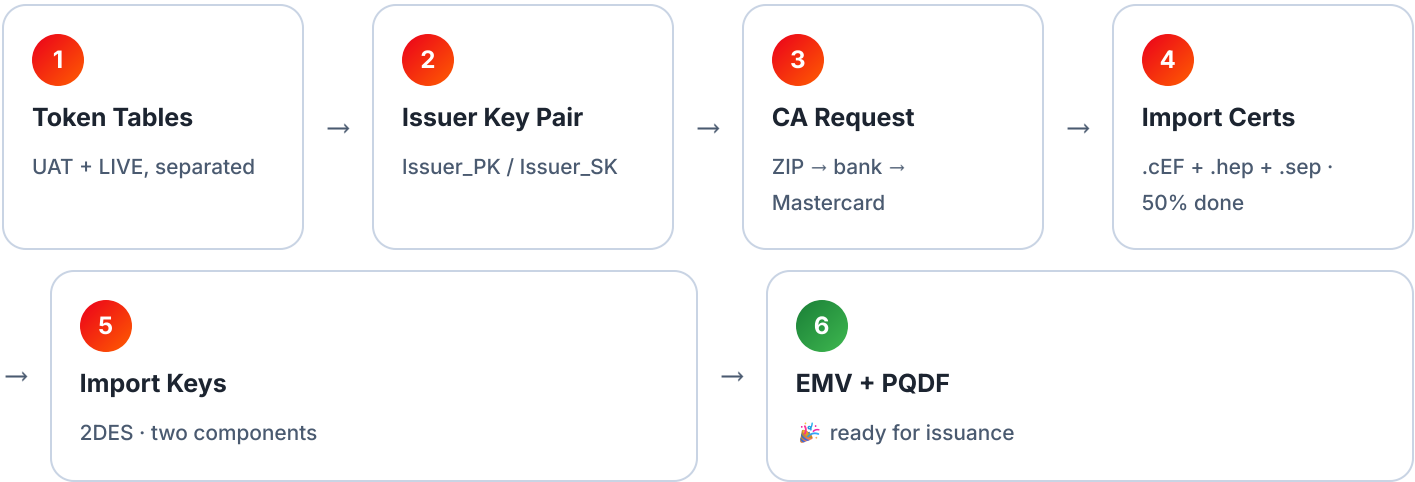
CA FAILURES FROM KEY
PAIR STEP

INTRODUCTION

Overview

Bringing a bank onto **Mastercard** or **Visa** rails using **Entrust** (nShield HSM + Key Manager) is mostly a *key-ceremony* discipline: the right keys, in the right token table, with the right attributes — followed by the right certificates, in the right order.

The full happy path



🕒 Roughly **50% of the mission is complete** once the CA and issuer certificates are imported. The remaining keys and EMV profile files finish the job.

PREREQUISITES



Before You Start — The Three Entrust Portals

Our mission uses **all three** portals in the Entrust ecosystem:

#	Portal	Primary Role in This Guide
1	 Key Manager Portal	Token tables, key pairs, CA requests, certificate & key imports
2	 EMV Profile Manager Portal	EMV kernel / profile configuration for the card applet
3	 Card Wizard Portal	Card personalization workflow & tool-file generation (PQDF)

Client-side prerequisites to confirm first

- ✓ Client maintains **separate token tables for Testing (UAT) and Live (PROD)** — see Step 1
- ✓ The bank's **BIN number** is available (it becomes the key *Owner*)
- ✓ A bank **HOST** operator is available to generate 2DES keys **with two key components**
- ✓ Access to the **Mastercard Portal** to download the `.hep` and `.sep` files

STEP 1 OF 6

Token Table Configuration

 Key Manager Portal · Setup → HSM Configurations

Some clients run **everything in one token table**. It is not a bad thing, but it is **not practical** — you want a clean blast-radius separation between testing and production material.

Rule of thumb: create **two separate token tables** — one for **Testing (UAT)**, one for **Live**.

To create a new token table

```
Entrust Key Manager Portal
├─ Setup
│   └─ HSM Configurations
│       └─ Right-click on the HSM → Login
│           └─ Create a New Token Table → Name the table
```

- ✓ When the table appears in the HSM tree, the first milestone is done.

STEP 2 OF 6 · MOST CRITICAL

Generate the Issuer Key Pair (Issuer_PK / Issuer_SK)

Next, generate the **Issuer Public Key (Issuer_PK)** — the key used to build the CA request — and the matching **Issuer Private Key (Issuer_SK)**.

⚠ This is the most critical step in the entire integration — roughly 80% of CA requests fail because of mistakes made here. Double-check every attribute before clicking *Generate*.

Navigation

```

Key Manager Portal
└─ Home
    └─ Choose the correct token table → Login
        └─ Login as Security Officer (SO)
            └─ "Generate Key Pair" (side bar) → Apply the parameters below
  
```

Required key parameters (Mastercard)

Parameter	Value	Notes
Key Type	Any RSA-based key type	Per your HSM's key-type naming
Key Length	1976 bits	As specified for Mastercard in this guide
Owner	The bank's BIN number	Ties the key material to the issuer identity
Version	06 = LIVE · EF = UAT	Version encodes the environment
Usages	Encrypt, Wrap, Verify	Select all three
Attributes	Extractable, Exportable, Deletable	Select all three
Exponent	03	Standard RSA public exponent

 **Generate both keys with the same procedure** — Issuer_PK and Issuer_SK follow the identical guide and identical parameters; only the **name** differs.

STEP 3 OF 6

Create & Submit the CA Request

 Key Manager Portal →  Mastercard Portal

Create the certificate authority request that will be submitted to Mastercard:

```
Key Manager Portal
├── Certificates
│   ├── Request a Certificate
│   │   ├── Mastercard
│   │   │   ├── Choose the Issuer Public Key created in Step 2
│   │   │   │   └── Generate Request Certificate → (ZIP output)
```

Hand-off flow

1.  Entrust produces the CA request as a **ZIP** archive.
2.  Give the ZIP to the **bank**; the bank submits it to **Mastercard**.
3.  Mastercard replies with the **Issuer Certificate** ( file).

 **Most banks mistakenly believe the .cEF is everything they need to go live with Mastercard. It is not.** You still need two additional files — both downloadable from the Mastercard Portal.

Artifact	Format	Where It Comes From	Used For
Issuer Certificate	<code>.cEF</code>	Mastercard reply to the CA request	Imported as the issuer certificate (Step 4)
Hardware Encryption Program	<code>.hep</code>	Mastercard Portal download	Building the CA certificate in Entrust (Step 4)
Security Program	<code>.sep</code>	Mastercard Portal download	Building the CA certificate in Entrust (Step 4)

✅ **Make sure you get all three:** `.cEF` + `.hep` + `.sep`.

STEP 4 OF 6 • THE 50% MILESTONE



Import the CA Certificate & Issuer Certificate



Key Manager Portal • Import Certificate

Once all three files are in hand, import them into Entrust **in this order**:

1 • Import the CA certificate (HEP + SEP)

```

Key Manager Portal
├─ Import Certificate
│   └─ CA Certificate
│       └─ Choose both .hep and .sep → Import

```

2 • Import the Issuer Certificate

```

Key Manager Portal
├─ Import Certificate
│   └─ Issuer Certificate
│       └─ Choose .cEF → Import

```

 At this point, 50% of the integration is complete.

STEP 5 OF 6

Import the Required Keys into the Token Table

 Key Manager Portal →  Bank HOST

The next phase is requesting and importing the **necessary operational keys** (request / session keys) into the token table.

 **Don't worry about guessing which keys you need** — Entrust will tell you exactly which keys must be configured. Once you have the list, pass it to the bank, and the bank generates them with its HOST.

 **Important instruction for the bank:** request **2DES keys generated with two components** — Entrust requires **at least two key components** to import a key.

Import procedure (once the components arrive)

```
Key Manager Portal
├─ Import Secret Key or Key Pair
│   └─ Import Key with Components
│       └─ Type in the plain key components
│           └─ Enter the components twice → Generate Key
```

STEP 6 OF 6 • FINAL MILE

EMV Profile & PQDF Tool Files

 EMV Profile Manager Portal •  Card Wizard Portal

After the keys are in place, Entrust provides:

Deliverable	Purpose
 EMV Profile file	Card applet / kernel configuration
 PQDF tool files	Personalization data tool files for card issuance

These are straightforward to import — simply follow the guidelines shipped with them.

 **Tuning tip:** expect to make small adjustments to the configuration files based on **your customer's card design and applet requirements** before the first production run.

SIGN-OFF

Delivery Checklist

Click items to tick them off — your progress is saved in this browser.

0 / 11 complete

- ☐ **Two token tables** exist — one **UAT**, one **LIVE**
- ☐ **Issuer_PK** & **Issuer_SK** generated with the correct BIN owner, version (**06** / **EF**), usages, attributes, and exponent **03**
- ☐ **CA request ZIP** generated and handed to the bank
- ☐ Bank submitted the request; Mastercard returned the **Issuer Certificate** (**.cEF**)
- ☐ **.hep** and **.sep** downloaded from the Mastercard Portal
- ☐ **CA certificate imported** (both **.hep** + **.sep**)
- ☐ **Issuer certificate imported** (**.cEF**)

☐ Required keys requested from the bank — **2DES, two components each**

☐ Keys imported via *Import Key with Components* (entered twice)

☐ **EMV profile** imported

☐ **PQDF tool files** imported & tuned for the customer's cards

LESSONS LEARNED



Common Pitfalls

#	Pitfall	Consequence	Avoidance
1	Wrong key-pair attributes (usages / attributes / exponent / version)	✗ ~80% of CA requests fail here	Re-read the Step 2 parameter table before generating
2	Thinking the <code>.cEF</code> alone is enough	Integration stalls at personalization	Remember: you need <code>.cEF</code> + <code>.hep</code> + <code>.sep</code>
3	One shared token table for UAT & Live	Operationally impractical; test material leaks into production risk	Always create separate tables (Step 1)
4	Bank generates keys as a single component	Entrust cannot import the key	Instruct the bank: 2DES keys with two components
5	Importing EMV/PQDF files without tuning	Cards misbehave for the specific customer design	Follow vendor guidelines and tune per customer cards



Happy issuing! 🎉

Written from hands-on field notes for payment-scheme engineers working with Entrust, Mastercard and Visa infrastructure.

Made with ❤️ by [MinThutaSawNaing](#) · [View on GitHub](#)